



US012417455B2

(12) **United States Patent**
Kcl et al.

(10) **Patent No.:** **US 12,417,455 B2**
(45) **Date of Patent:** ***Sep. 16, 2025**

(54) **SYSTEM AND METHOD FOR AUTOMATED LINKAGE OF ENRICHED TRANSACTION DATA TO A RECORD OF CHARGE**

(71) Applicant: **American Express Travel Related Services Company, Inc.**, New York, NY (US)

(72) Inventors: **Prasanth Kcl**, Phoenix, AZ (US);
Jordan Meyerowitz, Scottsdale, AZ (US); **Rajeev Thomas**, Phoenix, AZ (US)

(73) Assignee: **AMERICAN EXPRESS TRAVEL RELATED SERVICES COMPANY, INC.**, New York, NY (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/597,154**

(22) Filed: **Mar. 6, 2024**

(65) **Prior Publication Data**

US 2024/0289794 A1 Aug. 29, 2024

Related U.S. Application Data

(63) Continuation of application No. 18/166,315, filed on Feb. 8, 2023, now Pat. No. 11,954,682, which is a continuation of application No. 15/350,724, filed on Nov. 14, 2016, now Pat. No. 11,651,368.

(51) **Int. Cl.**
G06Q 20/40 (2012.01)
G06Q 20/38 (2012.01)

(52) **U.S. Cl.**
CPC **G06Q 20/401** (2013.01); **G06Q 20/382** (2013.01)

(58) **Field of Classification Search**
CPC G06Q 20/401; G06Q 20/382
USPC 705/75
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,811,810 B1	11/2017	Ranganath et al.
2003/0126094 A1	7/2003	Fisher et al.
2010/0082454 A1*	4/2010	Narayanaswami
		G06Q 30/0601
		705/26.1

2011/0131135 A1	6/2011	Carlson et al.
2011/0276441 A1	11/2011	Bond et al.
2012/0130897 A1	5/2012	Chisholm
2012/0166296 A1	6/2012	Hammond et al.

(Continued)

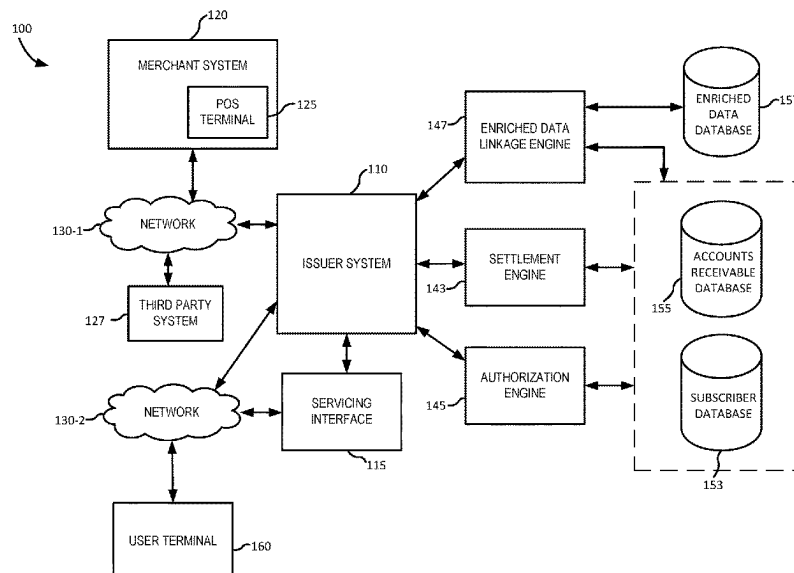
Primary Examiner — Courtney P Jones

(74) *Attorney, Agent, or Firm* — Thomas Horstemeyer, LLP

(57) **ABSTRACT**

Systems and methods for automated linkage of enriched transaction data to a record of charge are provided. Electronic enriched transaction data, such as receipt information, provided by merchants, merchant aggregators, or third parties may be automatically associated with the corresponding transaction, and transmitted after the sale, or after authorization of the transaction, without user intervention. The enriched transaction data may be automatically associated with the record of charge at the acquirer or issuer end. Transaction account holders are able to view and/or download the electronic enriched transaction data with the record of charge, without the need for manually initiating emailing or attaching or the electronic enriched transaction data.

20 Claims, 3 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2012/0239477	A1	9/2012	Cueli et al.	
2013/0268315	A1	10/2013	Cotton et al.	
2014/0040134	A1	2/2014	Peter	
2014/0040135	A1	2/2014	Ovick et al.	
2014/0046843	A1	2/2014	Liscia	
2014/0074675	A1 *	3/2014	Calman	G06Q 20/047 705/35
2014/0279311	A1	9/2014	Just	
2014/0310160	A1 *	10/2014	Kumar	G06Q 20/4016 705/39
2015/0032616	A1	1/2015	Blackhurst et al.	
2015/0032638	A1	1/2015	Dintenfass et al.	
2015/0073989	A1	3/2015	Green et al.	
2015/0142592	A1	5/2015	Chauhan	
2015/0206111	A1	7/2015	Sugiyama	
2015/0287077	A1	10/2015	Celikyilmaz et al.	
2015/0310564	A1	10/2015	Peterson	
2016/0132875	A1	5/2016	Blanco et al.	
2016/0148212	A1	5/2016	Dimmick	
2016/0171611	A1 *	6/2016	Sheng	G06Q 40/06 705/51
2016/0239839	A1	8/2016	Dillard et al.	
2016/0364959	A1	12/2016	Giddy et al.	
2017/0262784	A1	9/2017	Lowery et al.	
2019/0147429	A1	5/2019	Koh et al.	
2019/0236601	A1	8/2019	Izenson et al.	

* cited by examiner

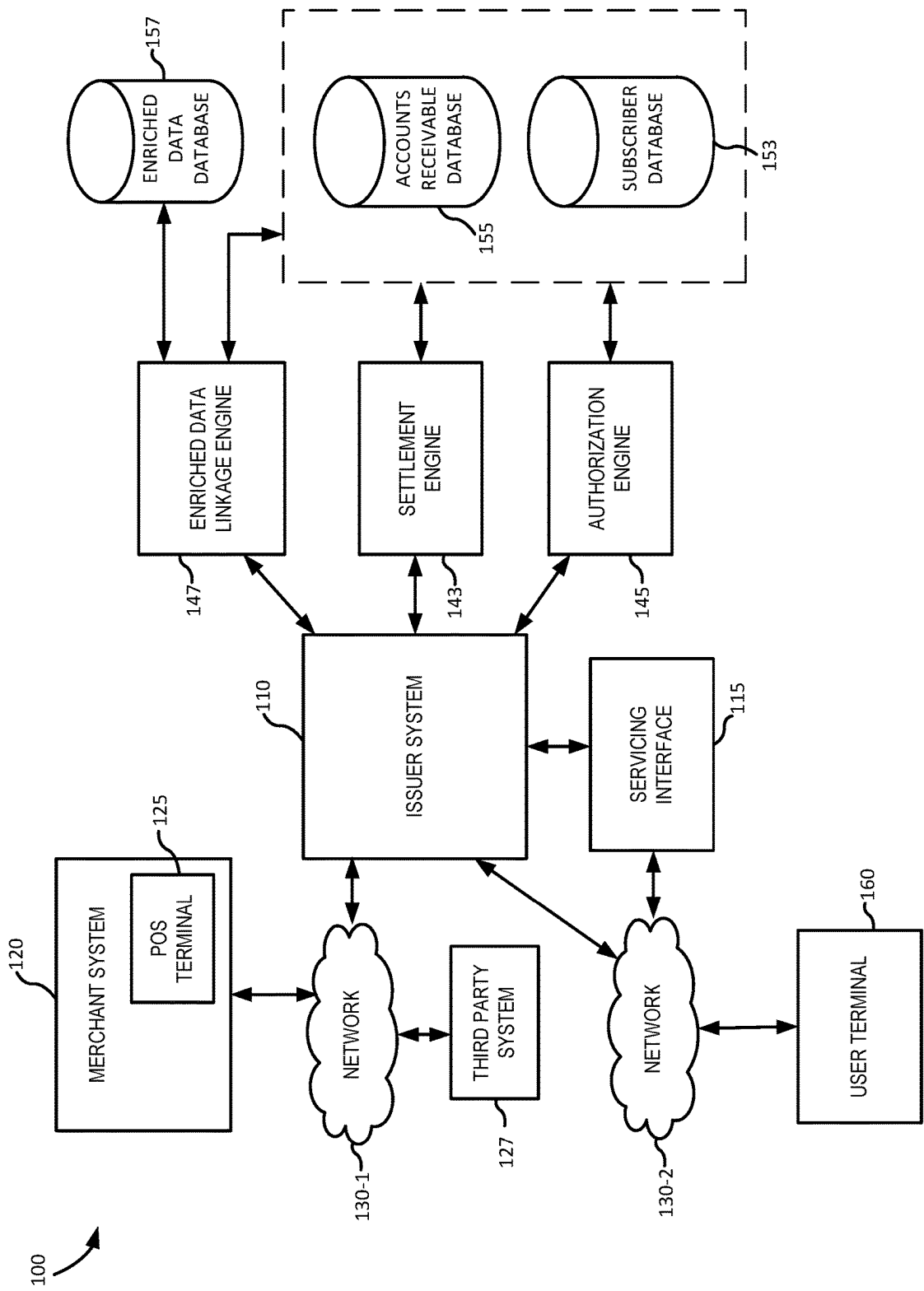
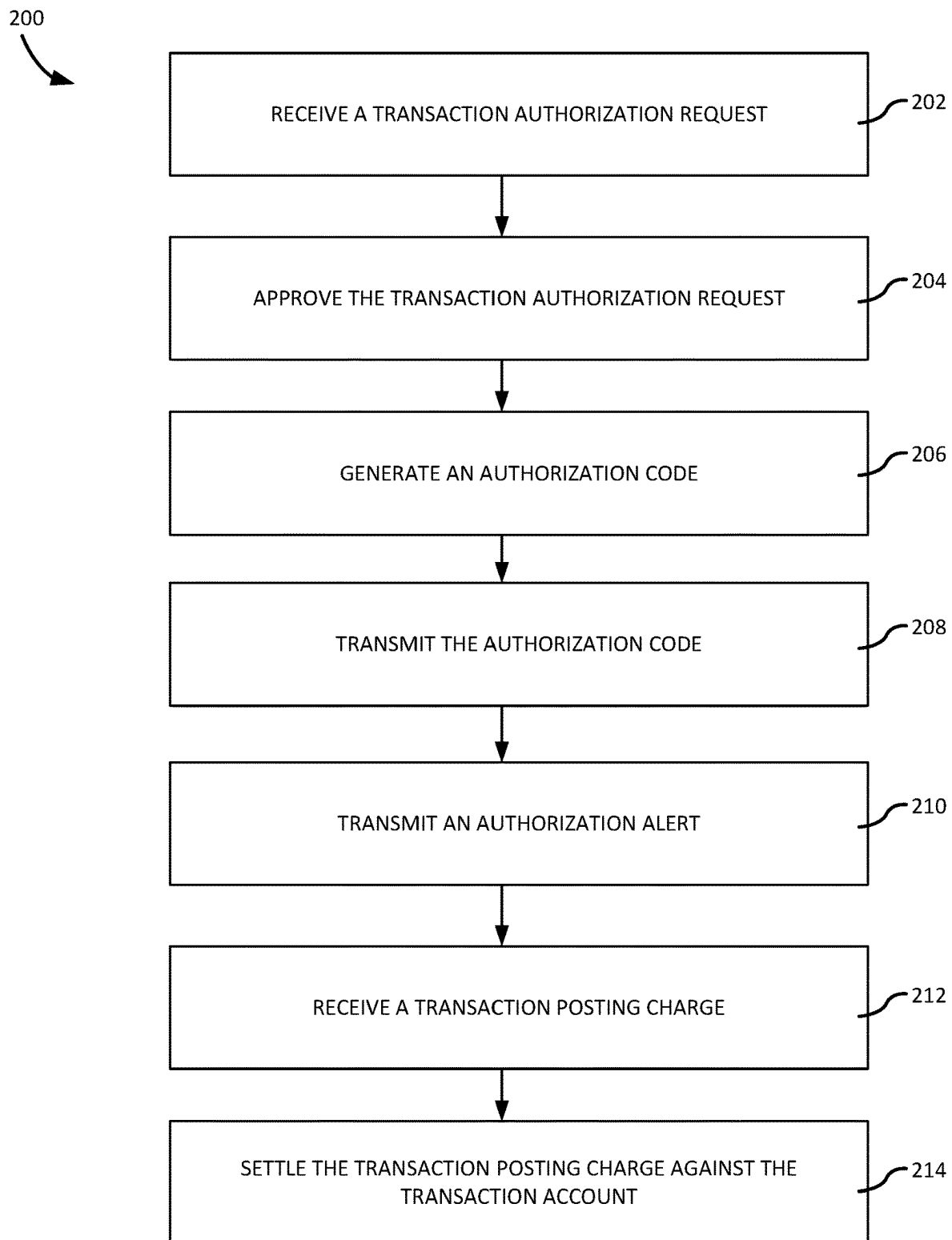
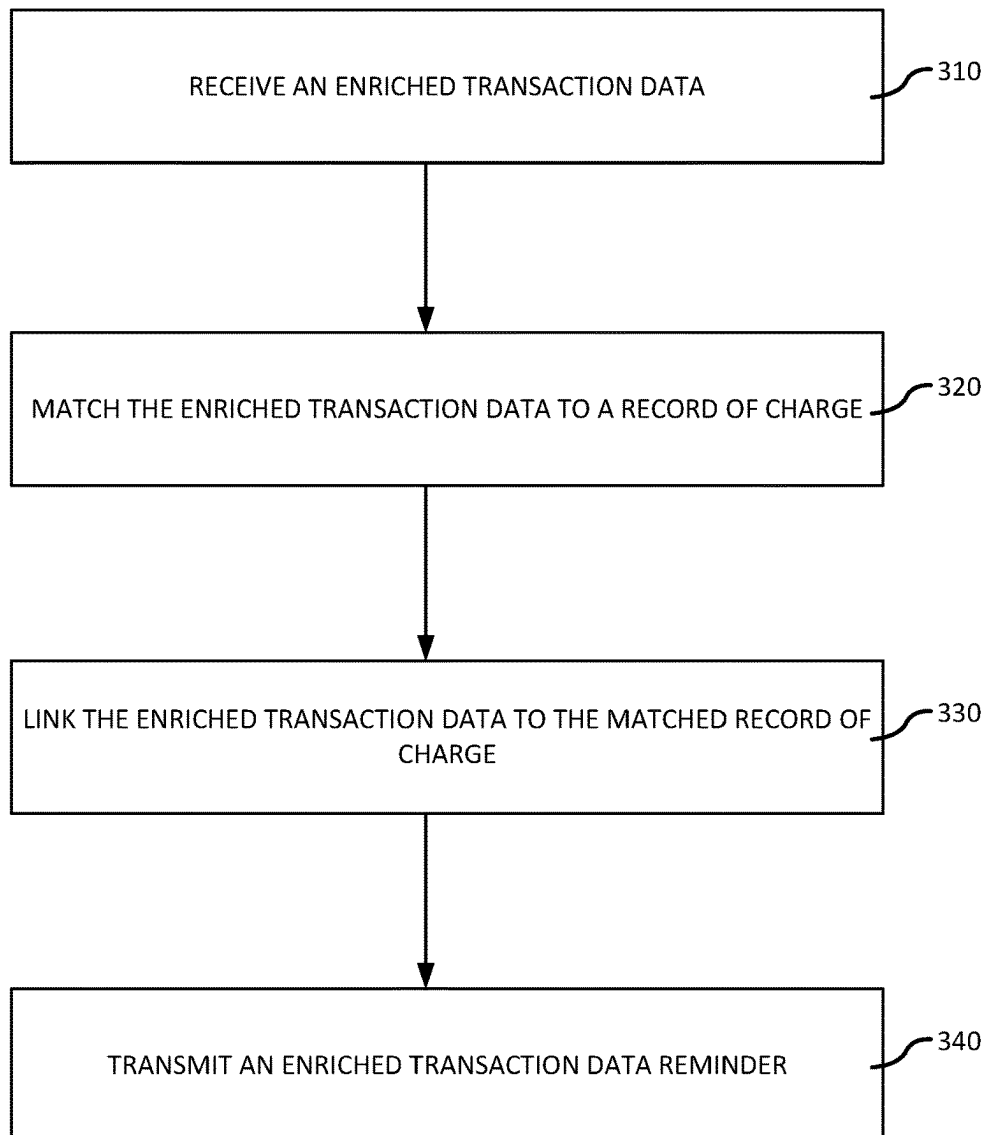


FIG. 1

**FIG. 2**

300

**FIG. 3**

1

SYSTEM AND METHOD FOR AUTOMATED LINKAGE OF ENRICHED TRANSACTION DATA TO A RECORD OF CHARGE

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of co-pending U.S. utility application entitled, “System and Method for Automated Linkage of Enriched Transaction Data to a Record of Charge,” having Ser. No. 18/166,315, filed Feb. 8, 2023, which is a continuation of U.S. utility application having Ser. No. 15/350,724, filed Nov. 14, 2016, all of the contents of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

The disclosure generally relates to financial transactions, and more specifically, to systems and methods for automated linkage of enriched transaction data to a record of charge at the acquirer or issuer end.

BACKGROUND

Credit cards, charge cards, and other transaction instruments may be commonly accepted today as a form of payment to a merchant, under a variety of circumstances. The transaction instrument may be used to complete a purchase in-person (e.g., at a retail store, a restaurant, a hotel, etc.), or may also be used to complete a purchase by relaying information associated with the transaction instrument (e.g., account number, account name, expiration data, billing address, etc.) to a merchant remotely, such as, for example, through the internet, by telephone, or by mail order. Transaction account owners may desire to view electronic receipt data, and other such enriched transaction data, related to each purchase. For example, a transaction account owner may desire to view electronic receipt data for individual purchases for convenience (e.g., effective expense management, reminders, simplified book keeping, etc.), to detect fraud, to dispute a purchase, and/or the like.

Transaction account owners may view transaction account statements, such as in an electronic and/or reoccurring statement, to review authorized (e.g., pending) charges and/or posted (e.g., settled) transactions. Typically, a transaction account owner receives a receipt, and other such enriched transaction data, as a written or electronic acknowledgment that a specified payment was received from multiple parties (e.g., from the merchant, a third party aggregator, a transaction account processing company, etc.) and manually sends the receipt, or other such enriched transaction data, or manually matches the enriched transaction data with each corresponding posted transaction from the transaction account statement.

SUMMARY

Systems, methods, and articles of manufacture (collectively, the “system”) for automated linkage of enriched transaction data are disclosed. The system may authorize a transaction between at least one of a merchant or a merchant aggregator and a transaction account holder. The system may transmit an authorization code to at least one of the merchant or the merchant aggregator. The authorization code may comprise data indicating that the transaction has been authorized and that a record of charge has been

2

generated. The system may receive enriched transaction data related to the transaction. The system may match the enriched transaction data to the record of charge. The system may associate the enriched transaction data to the record of charge.

In various embodiments, the enriched transaction data may be transmitted by at least one of the merchant, the merchant aggregator, or a third party. In various embodiments, the enriched transaction data may comprise at least one of electronic receipt information, a document, a tag, a note, a reminder related to the transaction, warranty information, a return policy, an itinerary, a deadline, a transaction location map, or a product manual. In various embodiments, the enriched transaction data may be matched to the record of charge based on at least one of a merchant ID, a merchant name, a transaction date, a transaction amount, an authorization code, or an authorization timestamp. In various embodiments, the system may further return the enriched transaction data to the transaction account holder, in response to a request for the enriched transaction data. In various embodiments, the system may return the enriched transaction data to the transaction account holder in response to a request for the enriched transaction data associated with the record of charge, wherein the enriched transaction data comprises an electronic receipt information. In various embodiments, the system may generate an enriched transaction data reminder based on the enriched transaction data. In various embodiments, the system may transmit the enriched transaction data reminder to the transaction account holder.

The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated herein otherwise. These features and elements as well as the operation of the disclosed embodiments will become more apparent in light of the following description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may be obtained by referring to the detailed description and claims when considered in connection with the drawing figures, wherein like numerals denote like elements.

FIG. 1 is a block diagram illustrating various system components of a system for automated linkage of enriched transaction data, in accordance with various embodiments;

FIG. 2 illustrates a process flow for authorizing transactions, in accordance with various embodiments; and

FIG. 3 illustrates a process flow for automated linkage of enriched transaction data to a record of charge, in accordance with various embodiments.

DETAILED DESCRIPTION

The detailed description of exemplary embodiments herein makes reference to the accompanying drawings, which show various embodiments by way of illustration. While these various embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that logical and mechanical changes may be made without departing from the spirit and scope of the disclosure. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation.

For example, the steps recited in any of the method or process descriptions may be executed in any order and are not limited to the order presented. Moreover, any of the functions or steps may be outsourced to or performed by one or more third parties. Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component may include a singular embodiment.

Phrases and terms similar to “financial institution” or “transaction account issuer” may include any entity that offers transaction account services. Although often referred to as a “financial institution,” the financial institution may represent any type of bank, lender or other type of account issuing institution, such as credit card companies, card sponsoring companies, or third party issuers under contract with financial institutions. It is further noted that other participants may be involved in some phases of the transaction, such as an intermediary settlement institution.

The terms “payment vehicle,” “financial transaction instrument,” “transaction instrument” and/or the plural form of these terms may be used interchangeably throughout to refer to a financial instrument. As used herein, an account code may or may not be associated with a physical financial instrument.

Phrases and terms similar to a “buyer”, “participant”, “consumer”, and “user” may include any entity that receives goods or services in exchange for consideration (e.g. financial payment). For example, a buyer may purchase, lease, rent, barter or otherwise obtain goods from a supplier and pay the supplier using a transaction account.

Phrases similar to a “processor” (e.g., payment processor) or “transaction account issuer” may include a company (e.g., a third party) appointed (e.g., by a merchant) to handle transactions. A payment processor may include an issuer, acquirer, authorizer and/or any other system or entity involved in the transaction process. Payment processors may be broken down into two types: front-end and back-end. Front-end payment processors have connections to various transaction accounts and supply authorization and settlement services to the merchant banks’ merchants. Back-end payment processors accept settlements from front-end payment processors and, via The Federal Reserve Bank, move money from an issuing bank to the merchant bank. In an operation that will usually take a few seconds, the payment processor will both check the details received by forwarding the details to the respective account’s issuing bank or card association for verification, and may carry out a series of anti-fraud measures against the transaction. Additional parameters, including the account’s country of issue and its previous payment history, may be used to gauge the probability of the transaction being approved. In response to the payment processor receiving confirmation that the transaction account details have been verified, the information may be relayed back to the merchant, who will then complete the payment transaction. In response to the verification being denied, the payment processor relays the information to the merchant, who may then decline the transaction.

As used herein, “transmit” may include sending electronic data from one system component to another over a network connection. Additionally, as used herein, “data” may include encompassing information such as commands, queries, files, data for storage, and the like in digital or any other form.

Phrases and terms similar to “transaction account” may include any account that may be used to facilitate a financial transaction. For example, a transaction account as used herein may refer to an account associated with an open account or a closed account system (as described herein) The transaction account may exist in a physical or non-physical

embodiment. For example, a transaction account may be distributed in non-physical embodiments such as an account number, frequent-flyer account, telephone calling account, and/or the like. Furthermore, a physical embodiment of a transaction account may be distributed as a financial instrument.

A record of charge (or “ROC”) may comprise any transaction or transaction data. The ROC may be a unique identifier associated with a transaction. A transaction may, in various embodiments, be performed by a one or more members using a transaction account, such as a transaction account associated with a gift card, a debit card, a credit card, and the like. A ROC may, in addition, contain details such as location, merchant name or identifier, transaction amount, transaction date, account number, account security pin or code, account expiration date, and the like for the transaction.

In various embodiments, transaction account authorizations may be performed at the transaction account issuer’s credit authorization system (“CAS”) in response to a swipe and/or submission of transaction instrument data by the transaction account holder at a Point of Sale (“PoS”) system and/or merchant. The CAS may issue an approval code as a proof of authorization. After the transaction authorization is complete, typically the merchant and/or PoS may provide a physical receipt and/or the like, containing enriched transaction data. Conventionally, a transaction account owner has to wait for the transaction to be posted by the merchant for the record of charge (ROC) to be available at a transaction account servicing website or mobile application (e.g., available electronically on a transaction account statement). However, the enriched transaction data is not linked to the corresponding record of charge, and is typically only manually linked, the efficiency of which may be dependent on optical character recognition (OCR), information quality, manual intervention, and/or the like. For additional information about enriched data and ROCs, see, for example, U.S. application Ser. No. 14/621,152, entitled “Automated Transfer of Enriched Transaction Account Data to a Submitted Record of Charge” and filed on Feb. 12, 2015, which is hereby incorporated by reference in its entirety for all purposes.

In various embodiments, the systems and methods provided herein may allow transaction account holders to access electronic receipts associated with a transaction authorization almost immediately (or at any time) after the transaction is complete. The systems and methods provided herein may enable automated linkage of electronic receipts to a transaction authorization done via any acquirer or processor. In this regard, more information is available at the transaction level, which may help to provide increased financial clarity to the transaction account holder. This may also result in easier accounting, book keeping, expense reporting, reduced disputes and/or chargebacks, merchant returns, and/or the like. Warranty documents, return policies, and tracking of services may also be accommodated by the system and methods described herein.

In various embodiments, enhanced satisfaction may be experienced by transaction account holders. For example, digital connection access may be provided to transaction account holders almost immediately (or at any time) after a transaction, which can be used for providing continuous support and services. Merchants may be able to provide additional offers to customers by leveraging the digital connection described herein. The systems and methods described herein may also provide for a reduction in disputes and a reduction in fraud, as a digital copy of documents,

5

receipts, and/or other enriched transaction data may be available in real time or soon after the time of the transaction. Receipt printing and/or point of sale maintenance costs may also be further reduced for merchants.

In various embodiments, a system **100** for automated linkage of enriched transaction data is disclosed. In various embodiments, system **100** may be computer based, and may comprise a processor, a tangible non-transitory computer-readable memory, and/or a network interface, along with other suitable system software and hardware components. Instructions stored on the tangible non-transitory memory may allow system **100** to perform various functions, as described herein. System **100** may also contemplate uses in association with web services, utility computing, pervasive and individualized computing, security and identity solutions, autonomic computing, cloud computing, commodity computing, mobility and wireless solutions, open source, biometrics, grid computing and/or mesh computing.

In various embodiments, system **100** may comprise an issuer system **110**. Issuer system **110** may be configured as a central hub to access various systems, engines, and components of system **100**. In that regard, issuer system **110** may comprise a network, computer-based system, and/or software component configured to provide an access point to various systems, engines, and components. Issuer system **110** may be in operative and/or electronic communication with various system **100** networks (e.g., network **130-1** and network **130-2**), a settlement engine **143**, an authorization engine **145**, and/or an enriched data linkage engine **147**. In various embodiments, issuer system **110** may also comprise a transaction account issuer's Credit Authorization System ("CAS") capable of authorizing transaction account purchase requests, as discussed further herein.

In various embodiments, system **100** may comprise a merchant system **120**. Merchant system **120** may incorporate hardware and/or software components. Merchant system **120** may be configured to enable a merchant to receive a payment from a user, and submit a request to authorize the payment. Merchant system **120** may comprise a POS terminal **125** configured as a mechanism to conduct a transaction. In this regard, POS terminal **125** may enable a user to initiate a transaction with a merchant. For example, POS terminal **125** may comprise a cashier station, a credit or debit card reader, and/or the like. POS terminal **125** may also comprise a near-field communication (NFC) terminal. In this regard, an NFC terminal may allow for the transfer of information to another NFC enabling device, such as, a mobile device. In various embodiments, POS terminal **125** may allow a user to pay a merchant using a virtual token stored on an NFC enabled mobile device.

In various embodiments, merchant system **120** and issuer system **110** may be interconnected via network **130-1** (e.g., a payment network). Network **130-1**, which may be part of certain transactions, represents existing proprietary networks that presently accommodate transactions for credit cards, debit cards, and/or other types of transactional instruments. Network **130-1** may be a closed network that is secure from eavesdroppers. In various embodiments, network **130-1** may comprise an exemplary transaction network such as American Express®, VisaNet®, Veriphone®, Discover Card®, PayPal®, ApplePay®, GooglePay®, private networks (e.g., department store networks), and/or any other payment networks.

As used herein, the term "network" may include any cloud, cloud computing system or electronic communications system or method which incorporates hardware and/or software components. Communication among the parties

6

may be accomplished through any suitable communication channels, such as, for example, a telephone network, an extranet, an intranet, Internet, point of interaction device (point of sale device, personal digital assistant (e.g., IPHONE®, BLACKBERRY®), cellular phone, kiosk, etc.), online communications, satellite communications, off-line communications, wireless communications, transponder communications, local area network (LAN), wide area network (WAN), virtual private network (VPN), networked or linked devices, keyboard, mouse and/or any suitable communication or data input modality. Moreover, although the system is frequently described herein as being implemented with TCP/IP communications protocols, the system may also be implemented using IPX, APPLE® talk, IP-6, Net-BIOS®, OSI, any tunneling protocol (e.g. IPsec, SSH), or any number of existing or future protocols. If the network is in the nature of a public network, such as the Internet, it may be advantageous to presume the network to be insecure and open to eavesdroppers. Specific information related to the protocols, standards, and application software utilized in connection with the Internet is generally known to those skilled in the art and, as such, need not be detailed herein.

The various system components may be independently, separately or collectively suitably coupled to the network via data links which includes, for example, a connection to an Internet Service Provider (ISP) over the local loop as is typically used in connection with standard modem communication, cable modem, Dish Networks®, ISDN, Digital Subscriber Line (DSL), or various wireless communication methods. It is noted that the network may be implemented as other types of networks, such as an interactive television (ITV) network. Moreover, the system contemplates the use, sale or distribution of any goods, services or information over any network having similar functionality described herein.

"Cloud" or "Cloud computing" includes a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. Cloud computing may include location-independent computing, whereby shared servers provide resources, software, and data to computers and other devices on demand. For more information regarding cloud computing, see the NIST's (National Institute of Standards and Technology) definition of cloud computing.

In response to a user initiating a transaction with a merchant system **120** through POS terminal **125**, merchant system **120** may send a transaction authorization request to authorization engine **145**, via issuer system **110**. The transaction authorization request may comprise transaction identification information, which may be standardized codes or numbers indicating the characteristics of the transaction (e.g. a merchant category code, a service establishment number, geographic location code, product code, POS terminal number, and/or the like), a transaction request amount, and/or a user profile identifier such as a transaction account number, a user account, a user account number, a card security code (CSC) or a card verification value (CVV), and/or any other identifying information.

In various embodiments, authorization engine **145** may be configured to authorize the transaction authorization request received from merchant system **120**. In that respect, authorization engine **145** may be in electronic and/or operative communication with a subscriber database **153**. Subscriber database **153** may store and maintain data regarding transaction accounts, such as, for example, a transaction account

number, user account access data (e.g. username, password, and/or the like), transaction account identifying information (e.g., owner address, city, state, zip code, etc.), a card security code (CSC) or a card verification value (CVV), transaction account balance, and/or other such similar data. Subscriber database 153 may store the data using any suitable technique described herein or known in the art. Authorization engine 145 may validate the risk of the transaction authorization request, such as, for example, by comparing the transaction authorization request to the stored transaction account data in subscriber database 153 (e.g., compare the transaction request amount to the transaction account balance, the CSV/CVV codes, owner identifying information, etc.). In response to approving the transaction authorization request, authorization engine 145 may transmit an authorization code, via issuer system 110, to merchant system 120. Merchant system 120 may store the authorization code as proof of the transaction and may issue a receipt to the transaction account owner as a written or electronic acknowledgment that a transaction payment was received.

In various embodiments, in response to authorizing the transaction authorization request, authorization engine 145 may communicate with an accounts receivable database 155 to store and maintain data on the authorized transactions. In that regard, accounts receivable database 155 may store the data as a pending transaction corresponding to the transaction account. Accounts receivable database 155 may store the data using any suitable technique described herein or known in the art. According to various embodiments, a pending transaction is an authorization that has taken place (e.g., the transaction is approved, and/or the goods have been exchanged for value), where the merchant (and/or a merchant aggregator) may decide to post the transaction for reimbursement (e.g., a posted transaction). For instance, there are many cases where the transaction may not be posted by the merchant. For example, hotel room reservations may often record a transaction account for the purpose of guaranty of available payment. Typically, at the end of the day (or at any other suitable time interval), a merchant, for the purpose of being reimbursed, will submit pending charges to an issuing bank and/or transaction account processor to convert the pending charge to a posted transaction (e.g., via settlement engine 143, as discussed further below). In this way, the pending charge is converted (settled) to a posted transaction.

In various embodiments, authorization engine 145 may also be configured to transmit a transaction authorization alert in response to authorizing the transaction authorization request. Authorization engine 145 may transmit the transaction authorization alert, via issuer system 110, to a user terminal 160. In that respect, authorization engine 145 may transmit the transaction authorization alert via email, SMS, MMS, smartphone notification, Facebook® Messenger, and/or the like. The transaction authorization alert may comprise data corresponding to the authorized transaction, such as, for example, the transaction request amount, merchant identifying information, and/or the like. In that respect, a transaction account owner may view the transaction authorization alert via user terminal 160.

In various embodiments, system 100 may comprise a settlement engine 143. In various embodiments, settlement engine 143 may be configured to process transaction information, convert pending charges into posted transactions, and settle posted transactions against a transaction account. Settlement engine 143 may process payment information by communicating with accounts receivable database 155 to

recall data on authorized transactions for a transaction account. Settlement engine 143 may also, in response to a successful payment transaction, send a transaction confirmation to merchant system 120, via issuer system 110, notifying a merchant of a successful payment transaction.

In various embodiments, system 100 may also comprise an enriched data linkage engine 147. Enriched data linkage engine 147 may be configured to receive enriched transaction data. In various embodiments, the enriched transaction data may comprise any suitable data related to a corresponding authorized and/or pending transaction. For example, the enriched transaction data may comprise electronic receipt information, additional detailed information, documents, tags, notes, reminders related to the transaction, warranty information, return policy, itineraries, reminders, deadlines, map with location of transaction, product manuals, and/or the like. Electronic receipt information may comprise data corresponding to the approved transaction, such as, for example, a merchant ID, a merchant name, a transaction date, a transaction amount, an authorization code, or an authorization timestamp. The enriched transaction data may be transmitted in any form, such as, for example, a URL, an image (e.g., .JPG, .BMP, .PNG, .TIF, .GIF, etc.), metadata, and/or the like.

In various embodiments, enriched data linkage engine 147 may receive the enriched transaction data from any suitable source, such as, for example, from merchant system 120 and/or third party system 127, via issuer system 110. In that respect enriched data linkage engine 147 may receive the enriched transaction data from a merchant, a merchant and/or payment aggregator (e.g., a provider that allows merchants to accept payments, such as Square®, PayPal®, and/or the like), a third party, and/or any other suitable party. Each party may interface with enriched data linkage engine 147 using any suitable technique. For example, a party may transmit the enriched transaction data through a submissions process in a transaction account processing system (e.g., via issuer system 110), through a receipt data capture interface, and/or the like. In various embodiments, enriched data linkage engine 147 may store the enriched transaction data in an enriched data database 157. Enriched data database 157 may store the enriched transaction data using any suitable technique.

For example, and in various embodiments, merchant system 120 may be configured to automatically transmit the enriched transaction data in response to receiving the authorization code from authorization engine 145.

In various embodiments, merchant system 120 may be configured to automatically transmit all received enriched transaction data. In that respect, merchant system 120 may transmit the enriched transaction data immediately, or at any other suitable time interval (e.g., merchant may submit batches of enriched transaction data), after generating a transaction receipt for a customer. In various embodiments, merchant system 120 may be configured to transmit all enriched transaction data, such as transaction receipts. Merchant system 120 may automatically transmit the enriched transaction data in response to receiving the authorization code (e.g., from authorization engine 145) and approving the transaction (e.g., after the charge is pending). Merchant system 120 may also automatically transmit the enriched transaction data in response to posting the charge, and/or at any other suitable time.

In various embodiments, enriched data linkage engine 147 may also receive the enriched transaction data from a third party system 127, via network 130-1 and issuer system 110. Third party system 127 may comprise any suitable third

party with access to enriched transaction data. For example, third party system 127 may comprise a receipt printer vendor for merchant system 120, such as, for example, Epson®, Star Micronics®, NCR®, Micros®, AGPtek®, and/or any other suitable receipt printer vendor. Third party system 127 may also comprise a data aggregator, and/or any other suitable third party. In various embodiments, third party system 127 may have a pre-connection with merchant system 120, via network 130-1, for example, such that third party system 127 may recognize the creation and/or receipt of enriched transaction data (e.g., merchant system 120 may communicate creation of enriched transaction data to third party system 127). In that respect, in response to a merchant generating (e.g., physically or electronically) a transaction receipt, third party system 127 may be configured to automatically capture data relating to the transaction receipt, and transmit the data to enriched data linkage engine 147, via network 130-1 and issuer system 110. In various embodiments, third party system 127 may be configured to transmit all enriched transaction data.

In various embodiments, enriched data linkage engine 147 may also be configured to match the enriched transaction data to a record of charge. In that respect, enriched data linkage engine 147 may search through accounts receivable database 155 for recent transactions to retrieve a record of charge matching the enriched transaction data. Enriched data linkage engine 147 may match the enriched transaction data based on any suitable factor. For example, enriched data linkage engine 147 may match the enriched transaction data based on transaction attributes, such as merchant ID, merchant name, transaction date, transaction amount, authorization code, authorization timestamp, and/or through any other suitable factor. In various embodiments, enriched data linkage engine 147 may search for an exact match, or may search for a partial match. For example, enriched data linkage engine 147 may search for an exact match of merchant ID, merchant name, authorization code, and/or similar factors, but may allow for some tolerance for transaction date (e.g., may be within one or two days), transaction amount, authorization timestamp, and/or the like. In various embodiments, enriched data linkage engine 147 may also match the enriched transaction data based on a unique transaction identifier and/or the like. In response to a record of charge being matched with the enriched transaction data, enriched data linkage engine 147 may transmit and/or incorporate the enriched transaction data into the corresponding record of charge (e.g., into the record of charge stored in accounts receivable database 155). In response to not locating an associated record of charge, enriched data linkage engine 147 may continue to search accounts receivable database 155 for a match. For example, enriched data linkage engine 147 may search through pending charges to locate a match for the enriched transaction data.

In various embodiments, enriched data linkage engine 147 may also be configured to transmit an enriched transaction data reminder. In that respect, enriched data linkage engine 147 may transmit the enriched transaction data reminder, via issuer system 110 and network 130-2, to user terminal 160. The enriched transaction data reminder may comprise a notification and/or the like, related to the enriched transaction data. For example, where the enriched transaction data comprises warranty information or return policy (e.g., a three year warranty or a thirty day return policy), the enriched transaction data reminder may be transmitted to remind a transaction account owner that the warranty or return policy is set to expire. In that regard, enriched data linkage engine 147 may transmit the enriched

transaction data reminder at a preset interval prior to the expiration of the warranty or return policy (e.g., one month, three months, six months, etc.). Enriched data linkage engine 147 may transmit the enriched transaction data reminder via email, SMS, MMS, smartphone notification, Facebook® Messenger, and/or the like. The user may view the enriched transaction data reminder via user terminal 160. In various embodiments, the user may interface with issuer system 110, via user terminal 160, to request to receive enriched transaction data reminders.

In various embodiments, system 100 may comprise a user terminal 160. User terminal 160 may be in operative and/or electronic communication with issuer system 110, and may be configured to interface with issuer system 110. In that respect, user terminal 160 may comprise any suitable hardware and/or software components capable of sending and receiving data. For example, user terminal 160 may comprise a personal computer, personal digital assistant, cellular phone, kiosk, and/or the like. User terminal 160 may also comprise a near-field communication (NFC) enabled device, such as a smartphone (e.g., IPHONE®, BLACKBERRY®, and/or the like), a smart-ring, and/or the like.

In various embodiments, user terminal 160 may be configured to allow a user access to transaction account information in issuer system 110. In that respect, issuer system 110 may comprise a servicing interface 115. Servicing interface 115 may comprise a website, application, and/or the like capable of allowing a user, via user terminal 160, access to issuer system 110. In that respect, servicing interface 115 may require a secure login (e.g., a username and password) in order to grant a user access to issuer system 110. For example, a transaction account owner may access issuer system 110, via servicing interface 115, by entering their secure login information to electronically view their corresponding transaction account, including transaction reports, records of charge, and/or the like. A transaction account owner may then select individual records of charge on an electronic transaction report. The transaction account owner may select each record of charge to view additional information, including the enriched transaction data corresponding to that record of charge.

For example, and in various embodiments, a transaction account owner may select an individual record of charge and select an option to view receipt data (e.g., the enriched transaction data) related to that record of charge. The receipt data may populate (e.g., as populated fields, as a virtual representation of the physical receipt, and/or the like), allowing the transaction account owner to view the enriched transaction data related to the record of charge. In that respect, the transaction account owner may view, download, and/or print the enriched transaction data, such as the receipt data, linked to each corresponding record of charge. For example, the transaction account owner may print the receipt data as a proof of the transaction, allowing the transaction account owner to submit the printed receipt data to a merchant for a refund, exchange, store credit, warranty purposes, and/or the like. In various embodiments, issuer system 110, via servicing interface 115, may also enable the transaction account owner to print the enriched transaction data in a secured and/or verified manner. In that respect, the enriched transaction data may print to comprise a verified marking, such as a watermark and/or the like, to aid against fraud and/or similar such acts.

Referring now to FIGS. 2 and 3 the process flows and screenshots depicted are merely embodiments and are not intended to limit the scope of the disclosure. For example, the steps recited in any of the method or process descriptions

11

may be executed in any order and are not limited to the order presented. It will be appreciated that the following description makes appropriate references not only to the steps and user interface elements depicted in FIGS. 2 and 3, but also to the various system components as described above with reference to FIG. 1.

In various embodiments, and with reference to FIG. 2, a method 200 for authorizing transactions is disclosed. In various embodiments, method 200 may comprise receiving a transaction authorization request (Step 202). Issuer system 110 may receive the transaction authorization request in response to a transaction account owner initiating a transaction with merchant system 120 (e.g., with a merchant and/or a merchant aggregator). The transaction authorization request may comprise transaction identification information, which may be standardized codes or numbers indicating the characteristics of the transaction (e.g., a merchant category code, a service establishment number, geographic location code, product code, POS terminal number, and/or the like), a transaction request amount, and/or a user profile identifier such as a transaction account number, a user account, a user account number, a card security code (CSC) or a card verification value (CVV), and/or any other identifying information.

In various embodiments, method 200 may comprise approving the transaction authorization request (Step 204). Issuer system 110, via authorization engine 145, may approve and authorize the transaction authorization request received from merchant system 120. Issuer system 110, via authorization engine 145, may validate the risk of the transaction authorization request, such as, for example, by comparing the transaction authorization request to the stored transaction account data in subscriber database 153. In response to validating the risk of the transaction authorization request, issuer system 110 may approve the transaction authorization request. In various embodiments, method 200 may comprise generating an authorization code (Step 206). In response to approving the transaction authorization request, issuer system 110, via authorization engine 145 may generate an authorization code as proof of the transaction. In response to approving the transaction authorization request, issuer system 110 may also generate a pending charge. In various embodiments, method 200 may comprise transmitting the authorization code (Step 208). Issuer system 110 may transmit the authorization code to merchant system 120 (e.g., to the merchant and/or the merchant aggregator).

In various embodiments, method 200 may also comprise transmitting an authorization alert (Step 210). Issuer system 110 may transmit the authorization alert to user terminal 160 via email, SMS, MMS, smartphone notification, Facebook® Messenger, and/or the like. The transaction authorization alert may comprise data corresponding to the authorized transaction, such as, for example, the transaction request amount, merchant identifying information, and/or the like.

In various embodiments, method 200 may comprise receiving a transaction posting charge (Step 212). Issuer system 110 may receive the transaction posting charge from merchant system 120. In various embodiments, method 200 may comprise settling the transaction posting charge against the transaction account (Step 214). Issuer system 110 may reconcile the transaction posting charge against the pending charge, and may settle the transaction against the transaction account. Settlement engine 143 may also, in response to a successful payment transaction, send a transaction confirmation to merchant system 120 to notify a merchant of a successful payment transaction.

12

In various embodiments, and with reference to FIG. 3, a method 300 for automated linkage of enriched transaction data to a record of charge is disclosed. In various embodiments, method 300 may comprise receiving an enriched transaction data (Step 310). Issuer system 110, via enriched data linkage engine 147, may be configured to receive the enriched transaction data. Issuer system 110 may receive the enriched transaction data from merchant system 120 and/or third party system 127. In various embodiments the enriched transaction data may comprise any suitable data related to a corresponding authorized and/or pending transaction. For example, the enriched transaction data may comprise electronic receipt information, additional detailed information, documents, tags, notes, reminders related to the transaction, warranty information, return policy, itineraries, deadlines, a map with the location of the transaction (e.g., a transaction location map providing details (such as address, geocode, etc.) regarding where the transaction took place), product manuals, and/or the like.

In various embodiments, method 300 may comprise matching the enriched transaction data to a record of charge (Step 320). Issuer system 110, via enriched data linkage engine 147, may be configured to match the enriched transaction data to a record of charge. Issuer system 110 may match the enriched transaction data based on transaction attributes, such as merchant ID, merchant name, transaction date, transaction amount, authorization code, authorization timestamp, and/or through any other suitable factor. In various embodiments, issuer system 110 may locate an exact match or a partial match. Issuer system 110 may search for recent transactions in accounts receivable database 155 to locate a record of charge matching the enriched transaction data. In various embodiments, method 300 may comprise linking the enriched transaction data to the matched record of charge (Step 330). In response to a record of charge being matched with the enriched transaction data, issuer system 110 may transmit the enriched transaction data into the corresponding record of charge (e.g., into the record of charge stored in accounts receivable database 155) to link the enriched transaction data to the corresponding record of charge. In that respect a transaction account owner may request to view, download, print, and/or the like the enriched transaction data linked to the corresponding record of charge, such as, for example, through viewing the record of charge on an online servicing system (as described in more detail herein). Issuer system 110 may return the enriched transaction data to the transaction account owner, via user terminal 160, in response to the transaction account owner requesting the enriched transaction data (e.g., accessing the record of charge and viewing, downloading, printing, and/or the like the enriched transaction data).

In various embodiments, method 300 may also comprise transmitting an enriched transaction data reminder (Step 340). Issuer system 110, via enriched data linkage engine 147, may transmit the enriched transaction data reminder to user terminal 160. The enriched transaction data reminder may comprise a notification, and/or the like, related to the enriched transaction data. Issuer system 110 may transmit the enriched transaction data reminder via email, SMS, MMS, smartphone notification, Facebook® Messenger, and/or the like, enabling a user to view the enriched transaction data reminder via user terminal 160.

The disclosure and claims do not describe only a particular outcome of linking an enriched transaction data to a record of charge, but the disclosure and claims include specific rules for implementing the outcome of linking an enriched transaction data to a record of charge and that

13

render information into a specific format that is then used and applied to create the desired results of linking an enriched transaction data to a record of charge, as set forth in *McRO, Inc v. Bandal Namco Ganes America Inc.* (Fed. Cir. case number 15-1080 Sep. 13, 2016). In other words, the outcome of linking an enriched transaction data to a record of charge can be performed by many different types of rules and combinations of rules, and this disclosure includes various embodiments with specific rules. While the absence of complete preemption may not guarantee that a claim is eligible, the disclosure does not sufficiently preempt the field of linking an enriched transaction data to a record of charge at all. The disclosure acts to narrow, confine, and otherwise tie down the disclosure so as not to cover the general abstract idea of just linking an enriched transaction data to a record of charge. Significantly, other systems and methods exist for linking an enriched transaction data to a record of charge, so it would be inappropriate to assert that the claimed invention preempts the field or monopolizes the basic tools of linking an enriched transaction data to a record of charge. In other words, the disclosure will not prevent others from linking an enriched transaction data to a record of charge, because other systems are already performing the functionality in different ways than the claimed invention. Moreover, the claimed invention includes an inventive concept that may be found in the non-conventional and non-generic arrangement of known, conventional pieces, in conformance with *Bascom v. AT&T Mobility*, 2015-1763 (Fed. Cir. 2016). The disclosure and claims go way beyond any conventionality of any one of the systems in that the interaction and synergy of the systems leads to additional functionality that is not provided by any one of the systems operating independently. The disclosure and claims may also include the interaction between multiple different systems, so the disclosure cannot be considered an implementation of a generic computer, or just “apply it” to an abstract process. The disclosure and claims may also be directed to improvements to software with a specific implementation of a solution to a problem in the software arts.

In various embodiments, the systems and methods may include alerting a subscriber when their computer is offline. With brief reference to FIG. 1, system 100 may include generating customized information, via issuer system 110, and alerting a remote subscriber that the information can be accessed from their computer. The alerts are generated by filtering received information, building information alerts and formatting the alerts into data blocks based upon subscriber preference information. The data blocks are transmitted to the subscriber’s wireless device (e.g., user terminal 160), which, when connected to the computer, causes the computer to auto-launch an application to display the information alert and provide access to more detailed information about the information alert. More particularly, the method may comprise providing a viewer application to a subscriber for installation on the remote subscriber computer; receiving information at a transmission server sent from a data source over the Internet, the transmission server comprising a microprocessor and a memory that stores the remote subscriber’s preferences for information format, destination address, specified information, and transmission schedule, wherein the microprocessor filters the received information by comparing the received information to the specified information; generates an information alert from the filtered information that contains a name, a price and a universal resource locator (URL), which specifies the location of the data source; formats the information alert into data blocks according to said information format; and transmits the

14

formatted information alert over a wireless communication channel to a wireless device associated with a subscriber based upon the destination address and transmission schedule, wherein the alert activates the application to cause the information alert to display on the remote subscriber computer and to enable connection via the URL to the data source over the Internet when the wireless device is locally connected to the remote subscriber computer and the remote subscriber computer comes online.

In various embodiments, the system and method may include a graphical user interface (e.g., via user terminal 160) for dynamically relocating/rescaling obscured textual information of an underlying window to become automatically viewable to the user. By permitting textual information to be dynamically relocated based on an overlap condition, the computer’s ability to display information is improved. More particularly, the method for dynamically relocating textual information within an underlying window displayed in a graphical user interface may comprise displaying a first window containing textual information in a first format within a graphical user interface on a computer screen; displaying a second window within the graphical user interface; constantly monitoring the boundaries of the first window and the second window to detect an overlap condition where the second window overlaps the first window such that the textual information in the first window is obscured from a user’s view; determining the textual information would not be completely viewable if relocated to an unobstructed portion of the first window; calculating a first measure of the area of the first window and a second measure of the area of the unobstructed portion of the first window; calculating a scaling factor which is proportional to the difference between the first measure and the second measure; scaling the textual information based upon the scaling factor, automatically relocating the scaled textual information, by a processor, to the unobscured portion of the first window in a second format during an overlap condition so that the entire scaled textual information is viewable on the computer screen by the user; and automatically returning the relocated scaled textual information, by the processor, to the first format within the first window when the overlap condition no longer exists.

In various embodiments, the system may also include isolating and removing malicious code from electronic messages (e.g., email) to prevent a computer (e.g., issuer system 110) from being compromised, for example by being infected with a computer virus. The system may scan, via issuer system 110 for example, electronic communications for malicious computer code and clean the electronic communication before it may initiate malicious acts. The system operates by physically isolating a received electronic communication in a “quarantine” sector of the computer memory. A quarantine sector is a memory sector created by the computer’s operating system such that files stored in that sector are not permitted to act on files outside that sector. When a communication containing malicious code is stored in the quarantine sector, the data contained within the communication is compared to malicious code-indicative patterns stored within a signature database. The presence of a particular malicious code-indicative pattern indicates the nature of the malicious code. The signature database further includes code markers that represent the beginning and end points of the malicious code. The malicious code is then extracted from malicious code-containing communication. An extraction routine is run by a file parsing component of the processing unit. The file parsing routine performs the following operations: scan the communication for the iden-

15

tified beginning malicious code marker; flag each scanned byte between the beginning marker and the successive end malicious code marker; continue scanning until no further beginning malicious code marker is found; and create a new data file by sequentially copying all non-flagged data bytes into the new file, which thus forms a sanitized communication file. The new, sanitized communication is transferred to a non-quarantine sector of the computer memory. Subsequently, all data on the quarantine sector is erased. More particularly, the system includes a method for protecting a computer from an electronic communication containing malicious code by receiving an electronic communication containing malicious code in a computer with a memory having a boot sector, a quarantine sector and a non-quarantine sector; storing the communication in the quarantine sector of the memory of the computer, wherein the quarantine sector is isolated from the boot and the non-quarantine sector in the computer memory, where code in the quarantine sector is prevented from performing write actions on other memory sectors; extracting, via file parsing, the malicious code from the electronic communication to create a sanitized electronic communication, wherein the extracting comprises scanning the communication for an identified beginning malicious code marker, flagging each scanned byte between the beginning marker and a successive end malicious code marker, continuing scanning until no further beginning malicious code marker is found, and creating a new data file by sequentially copying all non-flagged data bytes into a new file that forms a sanitized communication file; transferring the sanitized electronic communication to the non-quarantine sector of the memory; and deleting all data remaining in the quarantine sector.

In various embodiments, system **100** may also address the problem of retaining control over customers during affiliate purchase transactions, using a system for co-marketing the “look and feel” of the host web page with the product-related content information of the advertising merchant’s web page. System **100** can be operated by a third-party outsource provider, who acts as a broker between multiple hosts and merchants. Prior to implementation, a host places links to a merchant’s webpage on the host’s web page. The links are associated with product-related content on the merchant’s web page. Additionally, the outsource provider system stores the “look and feel” information from each host’s web pages in a computer data store, which is coupled to a computer server. The “look and feel” information includes visually perceptible elements such as logos, colors, page layout, navigation system, frames, mouse-over effects or other elements that are consistent through some or all of each host’s respective web pages. A customer who clicks on an advertising link, via user terminal **160**, for example, is not transported from the host web page to the merchant’s web page, but instead is re-directed to a composite web page that combines product information associated with the selected item and visually perceptible elements of the host web page. The outsource provider’s server responds by first identifying the host web page where the link has been selected and retrieving the corresponding stored “look and feel” information. The server constructs a composite web page using the retrieved “look and feel” information of the host web page, with the product-related content embedded within it, so that the composite web page is visually perceived by the customer as associated with the host web page. The server then transmits and presents this composite web page to the customer so that she effectively remains on the host web page to purchase the item without being redirected to the third party merchant affiliate. Because such composite pages

16

are visually perceived by the customer as associated with the host web page, they give the customer the impression that she is viewing pages served by the host. Further, the customer is able to purchase the item without being redirected to the third party merchant affiliate, thus allowing the host to retain control over the customer. This system enables the host to receive the same advertising revenue streams as before but without the loss of visitor traffic and potential customers. More particularly, the system may be useful in an outsource provider serving web pages offering commercial opportunities. The computer store containing data, for each of a plurality of first web pages, defining a plurality of visually perceptible elements, which visually perceptible elements correspond to the plurality of first web pages; wherein each of the first web pages belongs to one of a plurality of web page owners; wherein each of the first web pages displays at least one active link associated with a commerce object associated with a buying opportunity of a selected one of a plurality of merchants; and wherein the selected merchant, the outsource provider, and the owner of the first web page displaying the associated link are each third parties with respect to one other, a computer server at the outsource provider, which computer server is coupled to the computer store and programmed to: receive from the web browser of a computer user a signal indicating activation of one of the links displayed by one of the first web pages; automatically identify as the source page the one of the first web pages on which the link has been activated; in response to identification of the source page, automatically retrieve the stored data corresponding to the source page; and using the data retrieved, automatically generate and transmit to the web browser a second web page that displays: information associated with the commerce object associated with the link that has been activated, and the plurality of visually perceptible elements visually corresponding to the source page.

As used herein, “satisfy”, “meet”, “match”, “associated with” or similar phrases may include an identical match, a partial match, meeting certain criteria, matching a subset of data, a correlation, satisfying certain criteria, a correspondence, an association, an algorithmic relationship and/or the like. Similarly, as used herein, “authenticate” or similar terms may include an exact authentication, a partial authentication, authenticating a subset of data, a correspondence, satisfying certain criteria, an association, an algorithmic relationship and/or the like.

Terms and phrases similar to “associate” and/or “associating” may include tagging, flagging, correlating, using a look-up table or any other method or system for indicating or creating a relationship between elements such as, for example, (i) a transaction account and (ii) an item (e.g., offer, reward, discount) and/or digital channel. Moreover, the associating may occur at any point, in response to any suitable action, event, or period of time. The associating may occur at pre-determined intervals, periodic, randomly, once, more than once, or in response to a suitable request or action. Any of the information may be distributed and/or accessed via a software enabled link, wherein the link may be sent via an email, text, post, social network input and/or any other method known in the art.

The phrases consumer, customer, user, account holder, account affiliate, cardmember or the like shall include any person, entity, business, government organization, business, software, hardware, machine associated with a transaction account, buys merchant offerings offered by one or more merchants using the account and/or who is legally designated for performing transactions on the account, regardless

of whether a physical card is associated with the account. For example, the cardmember may include a transaction account owner, a transaction account user, an account affiliate, a child account user, a subsidiary account user, a beneficiary of an account, a custodian of an account, and/or any other person or entity affiliated or associated with a transaction account.

The customer may be identified as a customer of interest to a merchant based on the customer's transaction history at the merchant, types of transactions, type of transaction account, frequency of transactions, number of transactions, lack of transactions, timing of transactions, transaction history at other merchants, demographic information, personal information (e.g., gender, race, religion), social media or any other online information, potential for transacting with the merchant and/or any other factors.

The account number may be distributed and stored in any form of plastic, electronic, magnetic, radio frequency, wireless, audio and/or optical device capable of transmitting or downloading data from itself to a second device. A consumer account number may be, for example, a sixteen-digit account number, although each credit provider has its own numbering system, such as the fifteen-digit numbering system used by American Express. Each company's account numbers comply with that company's standardized format such that the company using a fifteen-digit format will generally use three-spaced sets of numbers, as represented by the number "0000 000000 00000". The first five to seven digits are reserved for processing purposes and identify the issuing bank, account type, etc. In this example, the last (fifteenth) digit is used as a sum check for the fifteen digit number. The intermediary eight-to-eleven digits are used to uniquely identify the consumer. A merchant account number may be, for example, any number or alpha-numeric characters that identify a particular merchant for purposes of account acceptance, account reconciliation, reporting, or the like.

In various embodiments, an account number may identify a consumer. In addition, in various embodiments, a consumer may be identified by a variety of identifiers, including, for example, an email address, a telephone number, a cookie id, a radio frequency identifier (RFID), a biometric, and the like.

The system may include or interface with any of the foregoing accounts, devices, and/or a transponder and reader (e.g. RFID reader) in RF communication with the transponder (which may include a fob), or communications between an initiator and a target enabled by near field communications (NFC). Typical devices may include, for example, a key ring, tag, card, cell phone, wristwatch or any such form capable of being presented for interrogation. Moreover, the system, computing unit or device discussed herein may include a "pervasive computing device," which may include a traditionally non-computerized device that is embedded with a computing unit. Examples may include watches, Internet enabled kitchen appliances, restaurant tables embedded with RF readers, wallets or purses with imbedded transponders, etc. Furthermore, a device or financial transaction instrument may have electronic and communications functionality enabled, for example, by: a network of electronic circuitry that is printed or otherwise incorporated onto or within the transaction instrument (and typically referred to as a "smart card"): a fob having a transponder and an RFID reader; and/or near field communication (NFC) technologies. For more information regarding NFC, refer to the following specifications all of which are incorporated by reference herein: ISO/IEC 18092/ECMA-340, Near Field

Communication Interface and Protocol-1 (NFCIP-1); ISO/IEC 21481/ECMA-352, Near Field Communication Interface and Protocol-2 (NFCIP-2); and EMV 4.2 available at <http://www.emvco.com/default.aspx>.

As used herein an "identifier" may be any suitable identifier that uniquely identifies an item. For example, the identifier may be a globally unique identifier ("GUID"). The GUID may be an identifier created and/or implemented under the universally unique identifier standard. Moreover, the GUID may be stored as 128-bit value that can be displayed as 32 hexadecimal digits. The identifier may also include a major number, and a minor number. The major number and minor number may each be 16 bit integers.

As used herein, big data may refer to partially or fully structured, semi-structured, or unstructured data sets including millions of rows and hundreds of thousands of columns. A big data set may be compiled, for example, from a history of purchase transactions over time, from web registrations, from social media, from records of charge ("ROC"), from summaries of charges ("SOC"), from internal data, or from other suitable sources. Big data sets may be compiled without descriptive metadata such as column types, counts, percentiles, or other interpretive-aid data points.

Distributed computing cluster may be, for example, a Hadoop® cluster configured to process and store big data sets with some of nodes comprising a distributed storage system and some of nodes comprising a distributed processing system. In that regard, distributed computing cluster may be configured to support a Hadoop® distributed file system (HDFS) as specified by the Apache Software Foundation at <http://hadoop.apache.org/docs/>. For more information on big data management systems, see U.S. Ser. No. 14/944,902 titled INTEGRATED BIG DATA INTERFACE FOR MULTIPLE STORAGE TYPES and filed on Nov. 18, 2015; U.S. Ser. No. 14/944,979 titled SYSTEM AND METHOD FOR READING AND WRITING TO BIG DATA STORAGE FORMATS and filed on Nov. 18, 2015; U.S. Ser. No. 14/945,032 titled SYSTEM AND METHOD FOR CREATING, TRACKING, AND MAINTAINING BIG DATA USE CASES and filed on Nov. 18, 2015; U.S. Ser. No. 14/944,849 titled SYSTEM AND METHOD FOR AUTOMATICALLY CAPTURING AND RECORDING LINEAGE DATA FOR BIG DATA RECORDS and filed on Nov. 18, 2015; U.S. Ser. No. 14/944,898 titled SYSTEMS AND METHODS FOR TRACKING SENSITIVE DATA IN A BIG DATA ENVIRONMENT and filed on Nov. 18, 2015; and U.S. Ser. No. 14/944,961 titled SYSTEM AND METHOD TRANSFORMING SOURCE DATA INTO OUTPUT DATA IN BIG DATA ENVIRONMENTS and filed on Nov. 18, 2015, the contents of each of which are herein incorporated by reference in their entirety.

Any communication, transmission and/or channel discussed herein may include any system or method for delivering content (e.g. data, information, metadata, etc.), and/or the content itself. The content may be presented in any form or medium, and in various embodiments, the content may be delivered electronically and/or capable of being presented electronically. For example, a channel may comprise a website or device (e.g., Facebook, YOUTUBE®, APPLE® TVR, PANDORA®, XBOX®, SONY® PLAYSTATION®), a uniform resource locator ("URL"), a document (e.g., a MICROSOFT® Word® document, a MICROSOFT® Excel® document, an ADOBE® .pdf document, etc.), an "ebook," an "emagazine," an application or microapplication (as described herein), an SMS or other type of text message, an email, Facebook® message, Twitter® tweet and/or message, MMS, and/or other type of

communication technology. In various embodiments, a channel may be hosted or provided by a data partner. In various embodiments, the distribution channel may comprise at least one of a merchant website, a social media website, affiliate or partner websites, an external vendor, a mobile device communication, social media network and/or location based service. Distribution channels may include at least one of a merchant website, a social media site, affiliate or partner websites, an external vendor, and a mobile device communication. Examples of social media sites include FACEBOOK®, FOURSQUARE®, TWITTER®, MYSPACE®, LINKEDIN®, and the like. Examples of affiliate or partner websites include AMERICAN EXPRESS®, GROUPON®, LIVINGSOCIAL®, and the like. Moreover, examples of mobile device communications include texting, email, and mobile applications for smart-phones.

In various embodiments, the methods described herein are implemented using the various particular machines described herein. The methods described herein may be implemented using the herein particular machines, and those hereinafter developed, in any suitable combination, as would be appreciated immediately by one skilled in the art. Further, as is unambiguous from this disclosure, the methods described herein may result in various transformations of certain articles.

For the sake of brevity, conventional data networking, application development and other functional aspects of the systems (and components of the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system.

The various system components discussed herein may include one or more of the following: a host server or other computing systems including a processor for processing digital data, a memory coupled to the processor for storing digital data; an input digitizer coupled to the processor for inputting digital data, an application program stored in the memory and accessible by the processor for directing processing of digital data by the processor; a display device coupled to the processor and memory for displaying information derived from digital data processed by the processor; and a plurality of databases. Various databases used herein may include: client data; merchant data, financial institution data; and/or like data useful in the operation of the system. As those skilled in the art will appreciate, user computer may include an operating system (e.g., WINDOWS®, OS2, UNIX®, LINUX®, SOLARIS®, MacOS, etc.) as well as various conventional support software and drivers typically associated with computers.

The present system or any part(s) or function(s) thereof may be implemented using hardware, software or a combination thereof and may be implemented in one or more computer systems or other processing systems. However, the manipulations performed by embodiments were often referred to in terms, such as matching or selecting, which are commonly associated with mental operations performed by a human operator. No such capability of a human operator is necessary, or desirable in most cases, in any of the operations described herein. Rather, the operations may be machine operations. Useful machines for performing the various embodiments include general purpose digital computers or similar devices.

In fact, in various embodiments, the embodiments are directed toward one or more computer systems capable of carrying out the functionality described herein. The computer system includes one or more processors, such as processor. The processor is connected to a communication infrastructure (e.g., a communications bus, cross over bar, or network) Various software embodiments are described in terms of this exemplary computer system. After reading this description, it will become apparent to a person skilled in the relevant art(s) how to implement various embodiments using other computer systems and/or architectures. Computer system can include a display interface that forwards graphics, text, and other data from the communication infrastructure (or from a frame buffer not shown) for display on a display unit.

Computer system may also include a main memory, such as for example random access memory (RAM), and may also include a secondary memory. The secondary memory may include, for example, a hard disk drive and/or a removable storage drive, representing a floppy disk drive, a magnetic tape drive, an optical disk drive, etc. The removable storage drive reads from and/or writes to a removable storage unit in a well-known manner. Removable storage unit represents a floppy disk, magnetic tape, optical disk, etc. which is read by and written to by removable storage drive. As will be appreciated, the removable storage unit includes a computer usable storage medium having stored therein computer software and/or data.

In various embodiments, secondary memory may include other similar devices for allowing computer programs or other instructions to be loaded into computer system. Such devices may include, for example, a removable storage unit and an interface. Examples of such may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an erasable programmable read only memory (EPROM), or programmable read only memory (PROM)) and associated socket, and other removable storage units and interfaces, which allow software and data to be transferred from the removable storage unit to computer system.

Computer system may also include a communications interface. Communications interface allows software and data to be transferred between computer system and external devices. Examples of communications interface may include a modem, a network interface (such as an Ethernet card), a communications port, a Personal Computer Memory Card International Association (PCMCIA) slot and card, etc. Software and data transferred via communications interface are in the form of signals which may be electronic, electromagnetic, optical or other signals capable of being received by communications interface. These signals are provided to communications interface via a communications path (e.g., channel). This channel carries signals and may be implemented using wire, cable, fiber optics, a telephone line, a cellular link, a radio frequency (RF) link, wireless and other communications channels.

The terms "computer program medium" and "computer usable medium" and "computer readable medium" are used to generally refer to media such as removable storage drive and a hard disk installed in hard disk drive. These computer program products provide software to computer system.

Computer programs (also referred to as computer control logic) are stored in main memory and/or secondary memory. Computer programs may also be received via communications interface. Such computer programs, when executed, enable the computer system to perform the features as discussed herein. In particular, the computer programs,

when executed, enable the processor to perform the features of various embodiments. Accordingly, such computer programs represent controllers of the computer system.

In various embodiments, software may be stored in a computer program product and loaded into computer system using removable storage drive, hard disk drive or communications interface. The control logic (software), when executed by the processor, causes the processor to perform the functions of various embodiments as described herein. In various embodiments, hardware components such as application specific integrated circuits (ASICs). Implementation of the hardware state machine so as to perform the functions described herein will be apparent to persons skilled in the relevant art(s).

In various embodiments, the server may include application servers (e.g. WEB SPHERE, WEB LOGIC, JBOSS, EDBR Postgres Plus Advanced Server® (PPAS), etc.). In various embodiments, the server may include web servers (e.g. APACHE, IIS, GWS, SUN JAVA® SYSTEM WEB SERVER).

A web client includes any device (e.g., personal computer) which communicates via any network, for example such as those discussed herein. Such browser applications comprise Internet browsing software installed within a computing unit or a system to conduct online transactions and/or communications. These computing units or systems may take the form of a computer or set of computers, although other types of computing units or systems may be used, including laptops, notebooks, tablets, hand held computers, personal digital assistants, set-top boxes, workstations, computer-servers, main frame computers, mini-computers, PC servers, pervasive computers, network sets of computers, personal computers, such as IPADS®, IMACS®, and MACBOOKS®, kiosks, terminals, point of sale (“POS”) devices and/or terminals, televisions, or any other device capable of receiving data over a network. A web-client may run MICROSOFT® INTERNET EXPLORER®, MOZILLA® FIREFOX®, GOOGLE® CHROME®, APPLE® Safari, or any other of the myriad software packages available for browsing the internet.

As those skilled in the art will appreciate that a web client may or may not be in direct contact with an application server. For example, a web client may access the services of an application server through another server and/or hardware component, which may have a direct or indirect connection to an Internet server. For example, a web client may communicate with an application server via a load balancer. In various embodiments, access is through a network or the Internet through a commercially-available web-browser software package.

As those skilled in the art will appreciate, a web client includes an operating system (e.g., WINDOWS® OS, OS2, UNIX® OS, LINUX® OS, SOLARIS®, MacOS, and/or the like) as well as various conventional support software and drivers typically associated with computers. A web client may include any suitable personal computer, network computer, workstation, personal digital assistant, cellular phone, smart phone, minicomputer, mainframe or the like. A web client can be in a home or business environment with access to a network. In various embodiments, access is through a network or the Internet through a commercially available web-browser software package. A web client may implement security protocols such as Secure Sockets Layer (SSL) and Transport Layer Security (“TLS”). A web client may implement several application layer protocols including http, https, ftp, and sftp.

In various embodiments, components, modules, and/or engines of system 100 may be implemented as micro-applications or micro-apps. Micro-apps are typically deployed in the context of a mobile operating system, including for example, a WINDOWS® mobile operating system, an ANDROID® Operating System, APPLE® IOS®, a BLACKBERRY® operating system and the like. The micro-app may be configured to leverage the resources of the larger operating system and associated hardware via a set of predetermined rules which govern the operations of various operating systems and hardware resources. For example, where a micro-app desires to communicate with a device or network other than the mobile device or mobile operating system, the micro-app may leverage the communication protocol of the operating system and associated device hardware under the predetermined rules of the mobile operating system. Moreover, where the micro-app desires an input from a user, the micro-app may be configured to request a response from the operating system which monitors various hardware components and communicates a detected input from the hardware to the micro-app.

Any databases discussed herein may include relational, hierarchical, graphical, blockchain, or object-oriented structure and/or any other database configurations. The databases may also include a flat file structure wherein data may be stored in a single file in the form of rows and columns, with no structure for indexing and no structural relationships between records. For example, a flat file structure may include a delimited text file, a CSV (comma-separated values) file, and/or any other suitable flat file structure. Common database products that may be used to implement the databases include DB2 by IBM® (Armonk, N.Y.), various database products available from ORACLE® Corporation (Redwood Shores, Calif.), MICROSOFT® Access® or MICROSOFT® SQL Server® by MICROSOFT® Corporation (Redmond, Wash.), MySQL by MySQL AB (Uppsala, Sweden), MongoDB®, Redis®, Apache Cassandra®, or any other suitable database product. Moreover, the databases may be organized in any suitable manner, for example, as data tables or lookup tables. Each record may be a single file, a series of files, a linked series of data fields or any other data structure.

The blockchain structure may include a distributed database that maintains a growing list of data records. The blockchain may provide enhanced security because each block may hold individual transactions and the results of any blockchain executables. Each block may contain a timestamp and a link to a previous block. Blocks may be linked because each block may include the hash of the prior block in the blockchain. The linked blocks form a chain, with only one successor block allowed to link to one other predecessor block.

Association of certain data may be accomplished through any desired data association technique such as those known or practiced in the art. For example, the association may be accomplished either manually or automatically. Automatic association techniques may include, for example, a database search, a database merge, GREP, AGREP, SQL, using a key field in the tables to speed searches, sequential searches through all the tables and files, sorting records in the file according to a known order to simplify lookup, and/or the like. The association step may be accomplished by a database merge function, for example, using a “key field” in pre-selected databases or data sectors. Various database tuning steps are contemplated to optimize database perfor-

mance. For example, frequently used files such as indexes may be placed on separate file systems to reduce In/Out (“I/O”) bottlenecks.

More particularly, a “key field” partitions the database according to the high-level class of objects defined by the key field. For example, certain types of data may then be designated as a key field in a plurality of related data tables and the data tables may be linked on the basis of the type of data in the key field. The data corresponding to the key field in each of the linked data tables is preferably the same or of the same type. However, data tables having similar, though not identical, data in the key fields may also be linked by using AGREP, for example. In accordance with one embodiment, any suitable data storage technique may be utilized to store data without a standard format. Data sets may be stored using any suitable technique, including, for example, storing individual files using an ISO/IEC 7816-4 file structure; implementing a domain whereby a dedicated file is selected that exposes one or more elementary files containing one or more data sets; using data sets stored in individual files using a hierarchical filing system; data sets stored as records in a single file (including compression, SQL accessible, hashed via one or more keys, numeric, alphabetical by first tuple, etc.); Binary Large Object (BLOB); stored as ungrouped data elements encoded using ISO/IEC 7816-6 data elements; stored as ungrouped data elements encoded using ISO/IEC Abstract Syntax Notation (ASN.1) as in ISO/IEC 8824 and 8825; and/or other proprietary techniques that may include fractal compression methods, image compression methods, etc.

In various embodiments, the ability to store a wide variety of information in different formats is facilitated by storing the information as a BLOB. Thus, any binary information can be stored in a storage space associated with a data set. As discussed above, the binary information may be stored in association with the system or external to but affiliated with the system. The BLOB method may store data sets as ungrouped data elements formatted as a block of binary via a fixed memory offset using fixed storage allocation, circular queue techniques, or best practices with respect to memory management (e.g., paged memory, least recently used, etc.). By using BLOB methods, the ability to store various data sets that have different formats facilitates the storage of data, in the database or associated with system, by multiple and unrelated owners of the data sets. For example, a first data set which may be stored may be provided by a first party, a second data set which may be stored may be provided by an unrelated second party, and yet a third data set which may be stored, may be provided by a third party unrelated to the first and second party. Each of these three exemplary data sets may contain different information that is stored using different data storage formats and/or techniques. Further, each data set may contain subsets of data that also may be distinct from other subsets.

As stated above, in various embodiments, the data can be stored without regard to a common format. However, the data set (e.g., BLOB) may be annotated in a standard manner when provided for manipulating the data in the database or system. The annotation may comprise a short header, trailer, or other appropriate indicator related to each data set that is configured to convey information useful in managing the various data sets. For example, the annotation may be called a “condition header”, “header”, “trailer”, or “status”, herein, and may comprise an indication of the status of the data set or may include an identifier correlated to a specific issuer or owner of the data. In one example, the first three bytes of each data set BLOB may be configured or configurable to

indicate the status of that particular data set; e.g., LOADED, INITIALIZED, READY, BLOCKED, REMOVABLE, or DELETED. Subsequent bytes of data may be used to indicate for example, the identity of the issuer, user, transaction/membership account identifier or the like. Each of these condition annotations are further discussed herein.

The data set annotation may also be used for other types of status information as well as various other purposes. For example, the data set annotation may include security information establishing access levels. It access levels may, for example, be configured to permit only certain individuals, levels of employees, companies, or other entities to access data sets, or to permit access to specific data sets based on the transaction, merchant, issuer, user or the like. Furthermore, the security information may restrict/permit only certain actions such as accessing, modifying, and/or deleting data sets. In one example, the data set annotation indicates that only the data set owner or the user are permitted to delete a data set, various identified users may be permitted to access the data set for reading, and others are altogether excluded from accessing the data set. However, other access restriction parameters may also be used allowing various entities to access a data set with various permission levels as appropriate.

The data, including the header or trailer may be received by a standalone interaction device configured to add, delete, modify, or augment the data in accordance with the header or trailer. As such, in one embodiment, the header or trailer is not stored on the transaction device along with the associated issuer-owned data but instead the appropriate action may be taken by providing to the user at the standalone device, the appropriate option for the action to be taken. The system may contemplate a data storage arrangement wherein the header or trailer, or header or trailer history, of the data is stored on the system, device, or transaction instrument in relation to the appropriate data.

One skilled in the art will also appreciate that, for security reasons, any databases, systems, devices, servers or other components of the system may consist of any combination thereof at a single location or at multiple locations, wherein each database or system includes any of various suitable security features, such as firewalls, access codes, encryption, decryption, compression, decompression, and/or the like.

Encryption may be performed by way of any of the techniques now available in the art or which may become available—e.g., Twofish, RSA, El Gamal, Schorr signature, DSA, PGP, PKI, GPG (GnuPG), and symmetric and asymmetric cryptosystems.

The computing unit of the web client may be further equipped with an Internet browser connected to the Internet or an intranet using standard dial-up, cable, DSL or any other Internet protocol known in the art. Transactions originating at a web client may pass through a firewall in order to prevent unauthorized access from users of other networks. Further, additional firewalls may be deployed between the varying components of CMS to further enhance security.

Firewall may include any hardware and/or software suitably configured to protect CMS components and/or enterprise computing resources from users of other networks. Further, a firewall may be configured to limit or restrict access to various systems and components behind the firewall for web clients connecting through a web server. Firewall may reside in varying configurations including Stateful Inspection, Proxy based, access control lists, and Packet Filtering among others. Firewall may be integrated within a web server or any other CMS components or may further reside as a separate entity. A firewall may implement

network address translation (“NAT”) and/or network address port translation (“NAPT”). A firewall may accommodate various tunneling protocols to facilitate secure communications, such as those used in virtual private networking. A firewall may implement a demilitarized zone (“DMZ”) to facilitate communications with a public network such as the Internet. A firewall may be integrated as software within an Internet server, any other application server components or may reside within another computing device or may take the form of a standalone hardware component.

The computers discussed herein may provide a suitable website or other Internet-based graphical user interface which is accessible by users. In one embodiment, the MICROSOFT® INTERNET INFORMATION SERVICES® (IIS), MICROSOFT® Transaction Server (“MTS”), and MICROSOFT® SQL Server, are used in conjunction with the MICROSOFT® operating system, MICROSOFT® web server software, a MICROSOFT® SQL Server database system, and a MICROSOFT® Commerce Server. Additionally, components such as Access or MICROSOFT® SQL Server, ORACLE®, Sybase, Informix MySQL, Interbase, etc., may be used to provide an Active Data Object (“ADO”) compliant database management system. In one embodiment, the Apache web server is used in conjunction with a Linux operating system, a MySQL database, and the Perl, PHP, Ruby, and/or Python programming languages.

Any of the communications, inputs, storage, databases or displays discussed herein may be facilitated through a website having web pages. The term “web page” as it is used herein is not meant to limit the type of documents and applications that might be used to interact with the user. For example, a typical website might include, in addition to standard HTML documents, various forms, JAVA® applets, JAVASCRIPT, active server pages (“ASP”), common gateway interface scripts (“CGI”), extensible markup language (“XML”), dynamic HTML, cascading style sheets (“CSS”), AJAX (Asynchronous JAVASCRIPT And XML), helper applications, plug-ins, and the like. A server may include a web service that receives a request from a web server, the request including a URL and an IP address (123.56.789.234). The web server retrieves the appropriate web pages and sends the data or applications for the web pages to the IP address. Web services are applications that are capable of interacting with other applications over a communications means, such as the internet. Web services are typically based on standards or protocols such as XML, SOAP, AJAX, WSDL and UDDI. Web services methods are well known in the art, and are covered in many standard texts. For example, representational state transfer (REST), or RESTful, web services may provide one way of enabling interoperability between applications.

Middleware may include any hardware and/or software suitably configured to facilitate communications and/or process transactions between disparate computing systems. Middleware components are commercially available and known in the art. Middleware may be implemented through commercially available hardware and/or software, through custom hardware and/or software components, or through a combination thereof. Middleware may reside in a variety of configurations and may exist as a standalone system or may be a software component residing on the Internet server. Middleware may be configured to process transactions between the various components of an application server and any number of internal or external systems for any of the purposes disclosed herein. WEBSPIHERE MQ™ (formerly

MQSenes) by IBM®, Inc. (Armonk, N.Y.) is an example of a commercially available middleware product. An Enterprise Service Bus (“ESB”) application is another example of middleware.

Those skilled in the art will also appreciate that there are a number of methods for displaying data within a browser-based document. Data may be represented as standard text or within a fixed list, scrollable list, drop-down list, editable text field, fixed text field, pop-up window, and the like. Likewise, there are a number of methods available for modifying data in a web page such as, for example, free text entry using a keyboard, selection of menu items, check boxes, option boxes, and the like.

The system and method may be described herein in terms of functional block components, screen shots, optional selections and various processing steps. It should be appreciated that such functional blocks may be realized by any number of hardware and/or software components configured to perform the specified functions. For example, the system may employ various integrated circuit components, e.g., memory elements, processing elements, logic elements, look-up tables, and the like, which may carry out a variety of functions under the control of one or more microprocessors or other control devices. Similarly, the software elements of the system may be implemented with any programming or scripting language such as C, C++, C#, JAVA®, JAVASCRIPT, JAVASCRIPT Object Notation (“JSON”), VBScript, Macromedia Cold Fusion, COBOL, MICROSOFT® Active Server Pages, assembly, PERL, PHP, awk, Python, Ruby, Visual Basic, SQL Stored Procedures, PL/SQL, any UNIX shell script, and extensible markup language (XML) with the various algorithms being implemented with any combination of data structures, objects, processes, routines or other programming elements. Further, it should be noted that the system may employ any number of conventional techniques for data transmission, signaling, data processing, network control, and the like. Still further, the system could be used to detect or prevent security issues with a client-side scripting language, such as JAVASCRIPT, VBScript or the like. Cryptography and network security methods are well known in the art, and are covered in many standard texts.

In various embodiments, the software elements of the system may also be implemented using Node.js®. Node.js® may implement several modules to handle various core functionalities. For example, a package management module, such as Npm®, may be implemented as an open source library to aid in organizing the installation and management of third-party Node.js® programs. Node.js® may also implement a process manager, such as, for example, Parallel Multithreaded Machine (“PM2”); a resource and performance monitoring tool, such as, for example, Node Application Metrics (“appmetrics”); a library module for building user interfaces, such as, for example, ReachJS®; and/or any other suitable and/or desired module.

As will be appreciated by one of ordinary skill in the art, the system may be embodied as a customization of an existing system, an add-on product, a processing apparatus executing upgraded software, a stand-alone system, a distributed system, a method, a data processing system, a device for data processing, and/or a computer program product. Accordingly, any portion of the system or a module may take the form of a processing apparatus executing code, an internet based embodiment, an entirely hardware embodiment, or an embodiment combining aspects of the internet, software and hardware. Furthermore, the system may take the form of a computer program product on a computer-

readable storage medium having computer-readable program code means embodied in the storage medium. Any suitable computer-readable storage medium may be utilized, including hard disks, CD-ROM, optical storage devices, magnetic storage devices, and/or the like.

The system and method is described herein with reference to screen shots, block diagrams and flowchart illustrations of methods, apparatus (e.g., systems), and computer program products according to various embodiments. It will be understood that each functional block of the block diagrams and the flowchart illustrations, and combinations of functional blocks in the block diagrams and flowchart illustrations, respectively, can be implemented by computer program instructions.

These computer program instructions may be loaded onto a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions that execute on the computer or other programmable data processing apparatus create means for implementing the functions specified in the flowchart block or blocks. These computer program instructions may also be stored in a computer-readable memory that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable memory produce an article of manufacture including instruction means which implement the function specified in the flowchart block or blocks. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the instructions which execute on the computer or other programmable apparatus provide steps for implementing the functions specified in the flowchart block or blocks.

Accordingly, functional blocks of the block diagrams and flowchart illustrations support combinations of means for performing the specified functions, combinations of steps for performing the specified functions, and program instruction means for performing the specified functions. It will also be understood that each functional block of the block diagrams and flowchart illustrations, and combinations of functional blocks in the block diagrams and flowchart illustrations, can be implemented by either special purpose hardware-based computer systems which perform the specified functions or steps, or suitable combinations of special purpose hardware and computer instructions. Further, illustrations of the process flows and the descriptions thereof may make reference to user WINDOWS®, webpages, websites, web forms, prompts, etc. Practitioners will appreciate that the illustrated steps described herein may comprise in any number of configurations including the use of WINDOWS®, webpages, web forms, popup WINDOWS®, prompts and the like. It should be further appreciated that the multiple steps as illustrated and described may be combined into single webpages and/or WINDOWS® but have been expanded for the sake of simplicity. In other cases, steps illustrated and described as single process steps may be separated into multiple webpages and/or WINDOWS® but have been combined for simplicity.

The term “non-transitory” is to be understood to remove only propagating transitory signals per se from the claim scope and does not relinquish rights to all standard computer-readable media that are not only propagating transitory signals per se. Stated another way, the meaning of the term “non-transitory computer-readable medium” and “non-transitory

sitory computer-readable storage medium” should be construed to exclude only those types of transitory computer-readable media which were found in *In Re Nuijten* to fall outside the scope of patentable subject matter under 35 U.S.C. § 101.

Systems, methods and computer program products are provided In the detailed description herein, references to “various embodiments”, “one embodiment”, “an embodiment”, “an example embodiment”, etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, and C” or “at least one of A, B, or C” is used in the claims or specification, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Although the disclosure includes a method, it is contemplated that it may be embodied as computer program instructions on a tangible computer-readable carrier, such as a magnetic or optical memory or a magnetic or optical disk. All structural, chemical, and functional equivalents to the elements of the above-described various embodiments that are known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the present claims. Moreover, it is not necessary for a device or method to address each and every problem sought to be solved by the present disclosure, for it to be encompassed by the present claims. Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is intended to be construed under the provisions of 35 U.S.C. 112 (f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises”, “comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

The invention claimed is:

1. A method comprising:

indicating, by at least one computing device of a transaction account issuer, that a transaction has been authorized between at least one of a merchant or a merchant aggregator and a transaction account holder and that a record of charge has been generated at a payment processor of the transaction account issuer by transmitting an authorization code to a point of sale terminal of the at least one of the merchant or the merchant aggregator;

in response to transmitting the authorization code, receiving, by the at least one computing device of the transaction account issuer, enriched transaction data to the record of charge stored in an accounts receivable database of the transaction account issuer from at least one of the merchant or the merchant aggregator;

linking, by the at least one computing device of the transaction account issuer, the enriched transaction data to the record of charge at the payment processor of the transaction account issuer by incorporating the enriched transaction data into the record of charge stored in the accounts receivable database of the transaction account issuer; and

transmitting, by the at least one computing device of the transaction account issuer, a user interface comprising the enriched transaction data.

2. The method of claim 1, wherein the enriched transaction data comprises an electronic receipt that is related to the transaction.

3. The method of claim 1, wherein the enriched transaction data comprises at least one of warranty documents, a product manual, or a uniform resource locator (URL) that provides access to a transaction location map that is related to the transaction.

4. The method of claim 1, further comprising matching, by the at least one computing device of the transaction account issuer, the enriched transaction data to the record of charge stored in the accounts receivable database of the transaction account issuer.

5. The method of claim 4, wherein the matching is based at least in part on at least one of an authorization code or an authorization timestamp associated with authorization of the transaction in response to receiving the enriched transaction data from at least one of the merchant or the merchant aggregator.

6. The method of claim 1, further comprising returning, by the at least one computing device of the transaction account issuer, the enriched transaction data to the transaction account holder, in response to a request for the enriched transaction data associated with the record of charge.

7. The method of claim 1, comprising generating, by the at least one computing device of the transaction account issuer, an enriched transaction data reminder based on the enriched transaction data, and transmitting, by the at least one computing device of the transaction account issuer, the enriched transaction data reminder to the transaction account holder.

8. A system comprising:

at least one computing device of a transaction account issuer; and

a tangible, non-transitory memory configured to communicate with a processor of the at least one computing device of the transaction account issuer, the tangible, non-transitory memory having instructions stored thereon that, in response to execution by the at least one computing device of the transaction account

issuer, cause the at least one computing device of the transaction account issuer to perform operations comprising:

indicating that a transaction has been authorized between at least one of a merchant or a merchant aggregator and a transaction account holder and that a record of charge has been generated at a payment processor of the transaction account issuer by transmitting an authorization code to a point of sale terminal of the at least one of the merchant or the merchant aggregator;

in response to transmitting the authorization code, receiving enriched transaction data to the record of charge stored in an accounts receivable database of the transaction account issuer from at least one of the merchant or the merchant aggregator;

linking the enriched transaction data to the record of charge at the payment processor of the transaction account issuer by incorporating the enriched transaction data into the record of charge stored in the accounts receivable database of the transaction account issuer; and

transmitting a user interface comprising the enriched transaction data.

9. The system of claim 8, wherein the enriched transaction data comprises an electronic receipt that is related to the transaction.

10. The system of claim 8, wherein the enriched transaction data comprises at least one of warranty documents, a product manual, or a uniform resource locator (URL) that provides access to a transaction location map that is related to the transaction.

11. The system of claim 8, wherein the instructions further cause the at least one computing device of the transaction account issuer to match the enriched transaction data to the record of charge stored in the accounts receivable database of the transaction account issuer.

12. The system of claim 11, wherein the matching is based at least in part on at least one of an authorization code or an authorization timestamp associated with authorization of the transaction in response to receiving the enriched transaction data from at least one of the merchant or the merchant aggregator.

13. The system of claim 8, wherein the instructions further cause the at least one computing device of the transaction account issuer to return the enriched transaction data to the transaction account holder, in response to a request for the enriched transaction data associated with the record of charge.

14. The system of claim 8, wherein the instructions further cause the at least one computing device of the transaction account issuer to generate an enriched transaction data reminder based on the enriched transaction data, and transmitting, by the at least one computing device of the transaction account issuer, the enriched transaction data reminder to the transaction account holder.

15. An article of manufacture including a non-transitory, tangible computer readable storage medium having instructions stored thereon that, in response to execution by a computer-based system of a transaction account issuer, cause the computer-based system of the transaction account issuer to perform operations comprising:

indicating, by the computer-based system of a transaction account issuer, that a transaction has been authorized between at least one of a merchant or a merchant aggregator and a transaction account holder and that a record of charge has been generated at a payment

31

processor of the transaction account issuer by transmitting an authorization code to a point of sale terminal of the at least one of the merchant or the merchant aggregator;

in response to transmitting the authorization code, receiving, by the computer-based system of a transaction account issuer, enriched transaction data to the record of charge stored in an accounts receivable database of the transaction account issuer from at least one of the merchant or the merchant aggregator;

linking, by the computer-based system of a transaction account issuer, the enriched transaction data to the record of charge at the payment processor of the transaction account issuer by incorporating the enriched transaction data into the record of charge stored in the accounts receivable database of the transaction account issuer; and

transmitting, by the computer-based system of a transaction account issuer, a user interface comprising the enriched transaction data.

16. The article of manufacture of claim **15**, wherein the enriched transaction data comprises an electronic receipt that is related to the transaction.

32

17. The article of manufacture of claim **15**, wherein the enriched transaction data comprises at least one of warranty documents, a product manual, or a uniform resource locator (URL) that provides access to a transaction location map that is related to the transaction.

18. The article of manufacture of claim **15**, wherein the instructions further cause the computer-based system of the transaction account issuer to match the enriched transaction data to the record of charge stored in the accounts receivable database of the transaction account issuer.

19. The article of manufacture of claim **18**, wherein the matching is based at least in part on at least one of an authorization code or an authorization timestamp associated with authorization of the transaction in response to receiving the enriched transaction data from at least one of the merchant or the merchant aggregator.

20. The article of manufacture of claim **15**, wherein the instructions further cause the computer-based system of the transaction account issuer to return the enriched transaction data to the transaction account holder, in response to a request for the enriched transaction data associated with the record of charge.

* * * * *